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Methods and Development

THIRD EDITION

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PART

I

General Principles

CHAPTER

1 Introduction

A Short History of Educational Computing

It has been about forty years since educators and computer scientists began using computers for instructional purposes. In that time span incredible advances have been made in computer technology and its availability. In the 1960s and most of the 1970s instructional computing took place on large mainframe computers or occasionally on medium-sized computers. Educational computing existed only at large universities and was largely restricted to reading and typing text. Developing instructional materials required learning computer programming, often in a low-level language unsuitable for the purpose.

The invention of the microcomputer near the end of the 1970s has brought about the rapid spread of computing in businesses, schools, and homes. Microcomputers have evolved from machines that depended on typing and text printouts and were difficult to program, to machines that allow interaction via text, graphics, voice, and pointing. The most recent developments in microcomputer technology provide even greater power and ease of use through advanced visual and auditory devices. They also permit networking many microcomputers to share information and resources. With all these advances has been a steady and dramatic decrease in cost.

In 1978 the first widely available microcomputers were released. The Apple II microcomputer succeeded in becoming the most common microcomputer for use in schools. Most early microcomputer courseware was designed for the Apple II, thus increasing its popularity in the schools even more.

The release of the IBM personal computer in 1981 resulted in a sudden expansion of the microcomputer market into business and industry. But the Apple II's early lead, lower cost, availability of courseware, and better integration of text, graphics, and color prevented the IBM-PC from penetrating the elementary and secondary school market.

The 1984 release of Apple's Macintosh computer instigated many changes in the field of microcomputing. This computer provided far better integration of text and graphics, better voice and music capability, and permitted not only typing for user input, but the mouse for pointing at and drawing on the screen. These and other features made the Macintosh

much easier to use than previous computers. But its cost, lack of color, and lack of courseware prevented it from having much initial impact on education. Its greatest impact, rather, was on the improved design of subsequent microcomputers, which copied its graphical power (and its overall ease of use), and put the mouse on a par with the keyboard.

Even though the Macintosh pioneered advances in usability, IBM-compatible computers rapidly gained the greater share of the desktop and laptop markets, especially as Microsoft improved its Windows operating system, which had many similarities to that of the Macintosh. The use of personal computers (PCs) continued to grow as companies developed and released systems that allowed PCs to communicate with each other. First, local area networks (LANs) were developed, which linked PCs that were physically close to each other. Then wide area networks (WANs) became available that linked LANs, as well as remote individual computers. The Internet, a worldwide collection of interconnected local and wide area networks, grew throughout the 1980s.

In the early 1990s came the development that has transformed the entire computing landscape, the creation of a part of the Internet known as the World Wide Web. Since the early 1990s, the Internet has transformed from a network used primarily by academia and government, for the exchange of textual materials, into a worldwide resource. Today, hundreds of millions of people use the Internet to pursue activities as diverse as shopping, dating, researching, forming associations, exchanging textual, graphic, and video information, and, of course, learning. The Internet has and will continue to transform everything we do.

Educational computing began with a few, large, government-funded projects on mainframe and minicomputers. The University of Illinois PLATO project began in 1960 (Alpert & Bitzer, 1970). The PLATO project eventually enabled computer-based instruction (CBI) to integrate text and graphics, and provided instructors with one of the first programming environments for computer-based instruction. Beginning in 1972, the Mitre Corporation's TICCIT (*T*ime-shared, *I*nteractive, *C*omputer-*C*ontrolled *I*nformation *T*elevision) project (Merrill, Schneider, & Fletcher, 1980) introduced computer-based instruction on minicomputers. With it came the concept of learner-controlled instruction (Wydra, 1980) and a particular philosophy for designing computer-based instruction, today known as *component design theory* (Merrill, 1983; 1987; 1988).

These early educational computing systems were sophisticated and had features similar to those available on the Web today. One of their drawbacks, however, was that communication and other costs were high. These costs, together with the advent of microcomputers, led to their demise and the takeover by desktop computers. In many ways, this transition set back the field of instructional computing because the many benefits of having user computers networked were lost. In addition, a number of years passed before microcomputer-based authoring software became available for developing instructional programs as sophisticated as were their mainframe predecessors.

The early days of instructional computing were filled with excitement and prophecies for the potential of great educational improvement through computer-based instruction. However, although there have been great strides in technology and availability, actual improvement in learning is less dramatic. In some ways, the Internet and World Wide Web have both advanced and hindered the growth of effective instructional software. They have advanced it by making computer networks easily accessible to so many people. They have hindered it by not providing tools for developing good instructional multimedia. At the time of writing (early 2000), software designed specifically for the

Web for developing instruction was not prevalent. HTML and Java, the two primary programming languages for the Web, were not designed for developing instructional applications, whereas traditional instructional authoring systems, such as Authorware, frequently caused difficulties due to their need for plug-ins—software additions needed by Web browsers to run these applications.

The current state of instructional computing is still in flux, with users facing issues of software and hardware incompatibility, as well as a lack of excellent and effective educational software. Two other factors hinder the success of instructional computing; a shortage of people skilled in developing quality courseware (which has resulted in much low-quality courseware) and disagreement within the field on how computers *should* be used in education. The availability of multimedia capabilities on so many computers has further confused the issues because many designers and developers believe that just adding multimedia elements makes software more instructionally effective.

Notwithstanding this confusion, using computers for training and education is growing rapidly, largely because of the popularity of the Internet and World Wide Web. We believe that the convenience of the Web causes people to both overlook the shortcomings of currently available software and fuels the demand for higher quality.

For the most part, documenting gains in learning is difficult when delivering instruction via computer. However, it is widely accepted that computer-based instruction at least reduces the time spent learning. Even if the learning itself is not better, reducing time is a benefit. Properly used, computers can improve learning effectiveness and efficiency (Christmann et al., 1997; Kulik & Kulik, 1991). In addition, using technology for learning has logistical benefits. Materials can be distributed more cheaply and easily; it is easier to ensure all users have the most recent version of the materials; learners can access the materials at their convenience; accessibility is facilitated for people with disabilities; and dangerous, expensive, or unique environments can be simulated to improve access.

The most obvious event of the last decade has been the explosion of the World Wide Web and its effect on learning with multimedia. But almost in parallel, this decade has witnessed a dramatic debate on paradigms for learning and instruction. Adherents of the new paradigm, constructivism, have argued against the older behavioral and instructivist approaches, and to some extent against the dominant cognitivist paradigm. Adherents of the traditional approaches, especially proponents of the Instructional Systems Design (ISD) approach, are equally critical of constructivist claims. This continuing debate is addressed frequently in this book. Something useful can be learned by studying these different approaches.

The field of instructional computing is still young and evolving. Progress has been made, but much remains to be learned regarding the best ways to harness the power of computers. The proliferation of educational and training applications on the Internet hopefully provide the momentum for people to take instructional multimedia more seriously. As this occurs, the amount of good materials for learning and instruction should increase.

■ When to Use the Computer to Facilitate Learning

Hundreds of research studies have been conducted to prove that using computers to teach is better than using books, teachers, films, or other more traditional methods. Overall, reviews of these studies claim a small effect in favor of computer-based instruction

(Kulik & Kulik, 1986; 1991). Some researchers have argued that small differences are either a research artifact or caused by some reason other than computer use (Clark, 1983), and considerable debate has surrounded whether computers or any medium can improve learning (Clark, 1994; Kozma, 1991, 1994; Tennyson, 1994). Researchers claiming the effect exists have argued that it would be greater but is artificially lowered because much of the instruction delivered via computer is poorly designed. Some constructivist educators argue that learning differences are hard to detect because most programs attempt to *teach* the learners rather than provide an environment in which learners construct their own meaningful and individualized learning. However, for another reason, we maintain that little difference is demonstrated by these many studies. Different media have different advantages. For teaching one topic (a particular aspect of reading comprehension, for example) with one set of learners (sixth graders), a book may be better. For another topic (operating a drill press) and different learners (factory workers), instructional video may be better.

If we were to chart out all the instructional topics, the wide variety of learners, and the many instructional situations, we would sometimes find an advantage for books, sometimes teachers, sometimes film or video, sometimes peer-tutoring, sometimes hands-on field experience, sometimes listening to an audiotape, and sometimes computers. Not surprisingly, across these many studies, which utilized a variety of topics, learners, and situations, little or no overall effect was found in favor of a single medium.

To take advantage of the computer's particular capabilities and not to waste them, our first rule for correctly using or developing instruction to be delivered via computer is to do so in situations where the computer is *likely* to be beneficial (Trollip & Alessi, 1988). These situations include those in which the cost of instruction by other methods is high (for example in military training); safety is a concern (chemistry laboratories); the material is hard to teach by other methods (graphing in calculus); extensive individual learner practice is needed (foreign language grammar and vocabulary); learner motivation is typically lacking (ancient world history); logistic difficulties exist in traditional instruction (science experiments that take a long time to complete); or the intended learners have special needs (such as visual or auditory disabilities that can be alleviated by multimedia allowing a choice between auditory and visual channels).

Although none of these situations guarantees that a computer will be beneficial as a vehicle for delivering instruction, they increase the probability of success. High quality and creative instructional design coupled with careful evaluation and revision are also necessary.

The situations above all fall within the broad area of providing *instruction.* That is, for the most part, content resides within the computer program and is made available to the learner in a variety of ways. Constructivist educators argue that we should focus on *learning*—that to put content in the computer and ask people to try to remember it misassigns learning roles. They argue that a person is better than a computer at organizing, reflecting on, and structuring information, which typically occurs in the computer in traditional instructional applications. On the other hand, the computer is better than a person at remembering information—typically the human's role in traditional computer-based instruction. Their approach is to use the computer as a tool for learning, where its role is to provide a "space" in which learning takes place, to store the results of learner activity, and to take care of time-consuming tasks that detract from learning (such as

complex calculations). We discuss these approaches to learning in Chapters 5 ("Hypermedia"), 7 ("Simulations"), 9 ("Tools and Open-Ended Environments") and 11 ("Web-Based Learning").

There is room and need for both approaches. In some situations, providing instruction is appropriate; in others, providing the tools for learning is appropriate. More often than not the two should be combined. Dogmatism on either side is unnecessary and unproductive.

The Process of Instruction

In the ongoing debate between constructivist educators and instructivist educators, instruction has been portrayed as an approach whereby knowledge is *given to* people, while learning is an approach whereby people *obtain* knowledge for themselves. Such a distinction is artificial and a misuse of semantics. The two, instruction and learning, almost always go hand in hand, and the term *instruction* should be understood much more generally. Instruction should be the *creation and use of environments in which learning is facilitated.* People certainly can learn completely on their own and without outside instigation. However, in institutions in which learning must be facilitated (such as public schools, universities, or businesses), instruction *is* that facilitation of learning, and it can combine a variety of approaches. Instruction can admittedly be strictly directed or be much more open-ended. Most instruction falls in between. However, to facilitate learning (i.e., to provide instruction) in at least a moderately efficient way, the process must include several essential activities. This section describes our model for successful instruction. According to that model, the following four activities or *phases of instruction* should occur for learning to be effective and efficient:

- Presenting information
- Guiding the learner
- Practicing
- Assessing learning

In contrast to this model, proponents of discovery or constructivist learning might omit the first phase or change the order of the first two phases. Research evidence in favor of discovery learning is usually limited to some learners (usually higher-ability learners) and some kinds of learning (such as problem solving). Our opinion is that discovery learning activities are most beneficial when placed within the context of a larger learning environment with these four activities. That is, guiding the learner can include discovery techniques. But as a complete model for building learning environments, discovery learning has not proved beneficial for most learners and most subject areas.

Presenting Information

The first three phases are based on research on successful classroom instruction (Rosenshine & Stevens, 1986). To teach something new, the instructor must first present information. This may be done in a number of ways. For verbal or pictorial information, an instructor may present rules and examples, show pictures, or provide other nonverbal

information. To teach skills, such as operating a 35-mm camera or doing long division, the instructor can model the skills to be learned. That is, the instructor actually performs the skills so that learners can imitate them.

An important method of presenting information is through example. Thus, in addition to stating the physical rule *force equals mass times acceleration,* the instructor can demonstrate applications of that physical rule, such as a truck accelerating more quickly when it is empty. The skill of long division can be modeled using a variety of numbers in the dividend and the divisor. Most learners require more than one example before they are able to apply a rule or skill.

Presentation of information can be accomplished with any medium, not just with live instructors. It often occurs completely under the control of the learner. When college students read their course textbooks, information is being presented. When factory workers watch videotapes about new equipment, information is being presented. When a learner does library research or engages in any activity to seek out existing knowledge or information, the first phase is occurring.

Guiding the Learner

The first phase, *presenting information,* is instructor or media centered. The second phase, *guiding the learner,* is more interactive and includes both the learner and the medium. For example, having observed a videotape presentation, the learner might now perform under instructor guidance. Again, this means different things depending on the nature of the material. The learner may answer questions about factual information, may apply rules and principles in problem-solving activities, or practice procedural skills. In each case, an instructor (or interactive medium) observes the learner, corrects errors, and gives suggestions or hints. If the learner distorts factual information, an instructor might remind the learner of the correct information, perhaps by repeating it. When the learner performs a skill incorrectly, a computer may model again the procedure or part of it. If a learner demonstrates misunderstanding of concepts or principles, a fellow learner might try to understand the confusion and dispel it.

In the classroom, guidance often takes the form of the instructor asking questions that learners must answer. When a question is answered incorrectly, the instructor may either tell the learner the correct answer or may ask leading questions to help the learner recall the correct information.

When one learns from a book, questions or suggested activities are sometimes included as guidance. But unlike in the classroom, if the learner does not do these correctly, true guidance does not occur. The learner may receive help only at some later time, such as when the instructor checks the learner's work and provides feedback.

Guidance is important in instruction because nobody learns everything from a single exposure. Learners make errors and are frequently unaware that they have. Learners must be made aware of these and correct them. The interactive process of the learner attempting to apply new knowledge, the instructor correcting and guiding, and the learner making further attempts are components frequently omitted in instruction and yet probably the most important.

As previously stated, not all models of teaching begin with the presentation of information. Discovery learning is based on the assumption that learners discover princi-

ples or develop skills through experimentation and practice. There is evidence that for some kinds of information, such as in the sciences or for the development of self-directed learning strategies, learner inquiry and discovery are effective (Derry and Lajoie, 1993; Jonassen, Peck, & Wilson, 1999; White, 1993). For the majority of regular school subjects and most procedural or physical skills, we regard a model that begins with the presentation of information as more efficient and demonstrably more successful (Klausmeier & Feldman, 1975; Koran, 1971; Merrill, 1974). In cases where a discovery approach is believed to have some advantage, we would stress that *guided* discovery (in contrast to undirected, free discovery) is more successful. The discovery activity should be a part of the guidance phase of instruction. It should follow some initial exposure to relevant material. And it usually should be followed by the other phases of instruction.

Practice

Learning is not complete when a learner can do something once or can demonstrate that he or she currently understands the material. The learner must usually be able to perform quickly or fluently, sometimes under conditions of distraction, with few or no errors. Furthermore, we usually want to learn information permanently rather than for a short time. Practicing a skill once or answering a single question does not guarantee retention. Repeated practice is often required to retain information and to become familiar with it.

The third phase, *practice,* is also learner centered. Although an instructor or interactive medium may observe the learner and makes corrections when errors are observed, the emphasis is on the learner practicing and the instructor making only short corrective statements.

Fluency and speed are related but slightly different aspects of well-learned information. To be fluent in a skill not only means doing it quickly, but doing it without thinking about it. To speak French fluently, for example, it is necessary that the correct words come automatically, without thinking. Reading, writing, spelling, arithmetic, driving a car, and countless other skills are almost worthless if not performed in this way.

On the other hand, some information does not require fluency. It does not matter whether one can perform a chemistry experiment or write a critical essay quickly. One need not be able to make decisions about starting a business without thinking about it. It is more important that such things be done carefully and correctly. However, one must at least remember how to do these things. Practice not only enhances speed and fluency but also retention.

Many examples of practice in classroom instruction exist. In elementary school reading instruction, the instructor frequently asks learners questions or requires them to read passages from primers. In arithmetic instruction, workbooks are the most common method of practice. They allow all learners to practice simultaneously rather than having most learners listen while one learner at a time practices. Unfortunately, when a learner makes an error practicing in a workbook, it might never be corrected.

In foreign-language instruction, a common type of practice is flashcards. The learner produces a pile of cards, for example, with French words on one side and equivalent English words on the reverse. The learner then goes through the deck of cards trying to translate the words correctly and receives immediate corrective feedback by looking at the other side of the card.

Assessing Learning

The first three phases just discussed are what most people consider to be instruction. However, we should not assume that instruction is successful for all learners. Rather, learning should be *assessed,* usually with tests or rubrics, which are an important part of the instructional process. These provide information about the level of learning, the quality of teaching, and future instructional needs. Instructors and learners alike place undue emphasis on assessment as a means of assigning grades. Our emphasis in this book is on assessment as a means of guiding instructional decisions—to determine what instruction is needed for which learners.

Methodologies for Facilitating Learning

According to the model we have described, the process of instruction includes the presentation of information to learners; guidance of learners' first interaction with the material; learners practicing the material to enhance fluency and retention; and, finally, assessment of learners to determine how well they have learned the material and what they should do next.

This model, though derived from research on successful classroom instruction, can also be applied to interactive multimedia. That is not to say that the computer must fulfill *all* the phases of instruction. Computers are but one element in a learning environment, along with teachers, other learners, and other media. The computer may serve one or a combination of the four phases. It may present initial information after which the learner receives guidance from an instructor and practices using a workbook. One may learn initial information from a lecture, after which the computer is used to practice some parts of the material for fluency. The computer may be used for the first three phases, with assessment being done in traditional ways. In all cases, the four phases of instruction should be present, generally using a combination of media.

When the computer *is* responsible for total instruction, it is important that all four phases be included. This is not always done. It is common, for example, for computer programs intended for practice (drills) to be expected to carry the load of total instruction. When this is attempted, learners can fail to learn what is desired.

In Part II of this book, we discuss eight methodologies of *Interactive Multimedia* (IMM) for the facilitation of learning:

- Tutorials
- Hypermedia
- Drills
- Simulations
- Games
- Tools and open-ended learning environments
- Tests
- Web-based learning

Tutorials are programs that generally engage the first two phases of instruction. They take the role of the instructor by presenting information and guiding the learner in initial acquisition. Hypermedia programs are another methodology for presenting or obtaining information but are designed for a more open-ended or constructivist learning experience. They are less structured than tutorials, thus allowing learners to choose their own paths through the material. Each individual would likely have different paths for different reasons.

Drills and most games typically engage learners in the third phase, helping them to practice for fluency and retention. We use the term *drill* for practice which *repeats* the material to be learned until it is mastered. Drill and game methodologies are often combined for motivational purposes. Some learning games are not repetitious, that is they do not practice to mastery, and so we would not call them drills, but they are nonetheless designed to provide practice in some area.

Simulations are more complicated. A simulation may be used to present information and guide the learner, to guide and practice, to do all three, or to assess a learner's knowledge. However, it is rare for a simulation (or a single lesson of *any* methodology) to provide all four phases of instruction. Most methodologies must be used in conjunction with other programs or media to provide complete instruction. The four phases of instruction typically occur over days or weeks, not a single instructional session.

Simulations may be used for direct instruction or for a more constructivist approach. Many simulations allow users to operate freely within a constrained environment. For example, in a program to simulate a chemistry titration experiment, the user may be allowed to assemble the equipment in a variety of ways or use any amounts of chemicals, but would not be able to use equipment or chemicals not shown. In a physics simulation for learning mechanics, the user could choose from a variety of devices for projecting or dropping objects and measuring their motions. Simulations may also be combined with the game methodology to foster discovery learning.

Games, as we have just said, may be combined with drills or with simulations. But many learning activities use games without being drills or simulations. Games may be used to practice information in a nonrepetitive manner, may be used as a discovery environment, or may be used to integrate learning across a number of subject areas, as is often done with the adventure game genre. Games usually support the third phase, practice, when they are combined with the drill methodology or used in the content integration fashion typical of adventure games. Less frequently, games may be used for guidance or assessment when combined with the simulation methodology.

Tools are computer software that learners use in conjunction with other media or activities for achieving some educational goal. They are by their nature more open ended and flexible. Graphics tools may be used to support drawing in art or graphing in math. Calculation tools may support science or business education. They may be a part of any of the phases of instruction and may support either constructivist or objectivist learning environments.

Open-ended learning environments, like simulations, provide an environment to support exploration. They usually include tool software as well. Although they may be used to foster learners obtaining or creating knowledge, they are frequently an environment in which learners practice the application of new knowledge.

Tests almost always represent the last phase, assessment of what has been learned. An exception is practice tests or quizzes, which are commonly used to foster the practice phase of instruction.

Finally, Web-based learning can be combined with any of these other methodologies (for the Web is essentially a *delivery* medium) though at this time, it is used mostly in conjunction with the hypermedia methodology. Use of the Web can foster any of the phases of instruction.

We devote a chapter to each of these eight approaches, and so it might be assumed that any interactive multimedia lesson must be classifiable as one of them. This is not the case. Many lessons combine methodologies, such as a lesson that begins with a tutorial and then follows with a drill, or a drill that is practiced in the context of a game to make it more enjoyable. In fact, it is rare for a program to use only one methodology. In the eight chapters of Part II, we describe and give examples of the methodologies, analyze their characteristics, and give recommendations for their design. The methodologies that we discuss in this book provide the basic groundwork for understanding and developing good interactive multimedia of either the instructivist or constructivist varieties.

Two Foundations of Interactive Multimedia

Before we analyze the methodologies of interactive multimedia, the next two chapters will provide a foundation on which to build the discussion. First, it is important to understand the different theories of learning that underlie all instruction and learning environments. Chapter 2 discusses those underlying issues.

Second, interactive multimedia must be built on sound human factors to be effective. Human factors is the study of the interaction of people with technology. Chapter 3 examines the general issues of screen design, multisensory presentation, types of interactions, learner control, and the facilitation of motivation.

Developing Interactive Multimedia

Finally, Part III discusses the various aspects of taking a design to fruition as a complete multimedia product. After a general introduction to development in Chapter 12, Chapter 13 provides a detailed look at how to plan a project properly, including the important issues of knowing the constraints under which you have to work (such as the budget), of keeping the project on track (project management), and of defining the overall standards for the project.

Chapter 14, "Design," discusses the process of defining the content, structure, and interactions of the multimedia program, and the essential issue of communicating your design ideas accurately to all team members.

Finally, Chapter 15, "Development," examines the development cycle and revisits project management. We consider production not only of the primary product but also of ancillary pieces, such as learner and instructor manuals. We also provide guidelines for testing your product and validating its effectiveness.

Conclusion

It is useful to lay out our own philosophy with respect to using computers to facilitate learning. First, we believe that there is a real world out there (an objectivist belief) and that people must learn to function appropriately in that world to survive and be successful. Second, we believe that learning is a constructive process whereby each learner observes and interprets reality and creates an understanding of it. Third, we believe that the two points above demand a combination of approaches—recognizing that learning is constructive, but also that there are essential truths to be learned. The fact that learning is constructive does not require that education be constructive. Rather, education should include direct instruction methods, experiential methods, exploration methods, and others.

Designing software that helps people to learn is a difficult and rewarding experience, no matter which philosophical approach you ascribe to. All philosophies share the common goal of having the people learn something useful and meaningful. We encourage you to jump right in and get your own interactive multimedia project underway (a constructivist approach) because you will learn much from the experience. Before you do so (or at least in parallel), we recommend you finish reading this book (a cognitivist approach) because it can save you a great deal of time and anguish by pointing out both well-established good practices, as well as known pitfalls.

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